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ComNets
Deutsche Telekom Chair
of Communication Networks

Faculty of Electronic and Information Institute of Communication Engineering

Chair of Communication Networks

Master Thesis

Randomized Identification Codes with a Focus on K-Identification

Chongyang Zheng

Born on: 10th October 1997 in Xi'an

Course: Nanoelectronic Systems

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Supervisors

Juan A. Cabrera

Caspar von Lengerke

Supervising professor

Prof. Dr. h.c. Fitzek



Task for Master Thesis of

Mr. Chongyang Zheng

Academic year: ET/

, matriculation number:

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Task description:

Identification via channels (ID) asks receivers to decide whether a specific message was sent, rather than to reconstruct it. Randomized ID codes achieve doubly-exponential message sets in blocklength and enable vanishing Type-I (false-accept) and Type-II (miss) error probabilities at positive ID rate. K-Identification (K-ID) generalizes this: the receiver asks whether the transmitted message belongs to a set of K candidates. K-ID is relevant for goal/semantics-driven communication (e.g., digital twins), where the action depends only on whether the current state falls into a small set of actionable hypotheses, not on exact message recovery. This thesis will study randomized ID codes with emphasis on K-ID, and extend the ideas and evaluation from “Stopping the Data Flood: Post-Shannon traffic reduction in digital-twins applications” to the K-ID setting. The work should balance theory, algorithms, and software-driven experiments using our provided library.

The student is expected to:

Survey the state of the art:

- Core ID concepts: randomized encoders/acceptance regions, ID capacity, error exponents, and operational interpretations; contrast with Shannon transmission.
- Define K-ID precisely (membership in a K-set) and position it vs. related notions (multiple-object ID, generalized ID; deterministic vs randomized variants).

Formulate the problem & metrics for K-ID:

- Formal model (channel class, randomization/common randomness budget, decoder tests).
- Report both Type-I/Type-II errors and computational metrics (encode/decode time, seed length, memory), plus finite-blocklength behavior.

Replicate & extend the digital-twins baseline to K-ID:

- Reproduce key experiments/results from the “Stopping the Data Flood” study (baseline ID) with our software stack.
- Design K-ID experiments that reflect digital-twin quantifying traffic reduction vs. reliability/latency.

Use our software library to build a modular K-ID test bench:

- Implement randomized ID/K-ID encoders and decoders (acceptance region tests) as pluggable components.

Design and run a simulation campaign:

- Measure KPIs such as: Type-I/II error curves, empirical exponents, queries per symbol, and compute/memory usage. Quantify traffic reduction vs. reliability/latency

Deliver a comparative assessment with clear takeaways:

- When do systems benefit from K-ID?
- Tradeoffs among randomization, complexity, and reliability.

Supervisors: Juan A. Cabrera, Caspar von Lengerke

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Prof. Dr.-Ing. Dr. h.c. Fitzek
Responsible Tutor



Statement of authorship

I hereby certify that I have authored this document entitled *Randomized Identification Codes with a Focus on K-Identification* independently and without undue assistance from third parties. No other than the resources and references indicated in this document have been used. I have marked both literal and accordingly adopted quotations as such. During the preparation of this document I was only supported by the following persons:

Juan A. Cabrera
Caspar von Lengerke

Additional persons were not involved in the intellectual preparation of the present document. I am aware that violations of this declaration may lead to subsequent withdrawal of the academic degree.

Dresden, 15th January 2026

Chongyang Zheng



Abstract

Identification via channels (ID) asks a receiver to decide whether a specific hypothesis about the transmitted message is true, rather than reconstructing the message itself. This document provides precise definitions, key capacity results, error metrics, and an experimental plan tailored to K-Identification (K-ID), where the receiver asks whether the message belongs to a set of K candidates. It contrasts randomized vs. deterministic (DI) variants, formulates problem and metrics, and outlines a modular test bench and simulation campaign aligned with the provided software stack.

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Acronyms

CPS	Cyber-Physical Systems	1
DT	Digital Twin	1–5
HRC	Human-Robot Collaboration	5
ID	Identification	1–3, 5
K-ID	K-Identification	1–3, 5
RS-ID	Reed-Solomon Identification	2–4
SHA-256	Secure Hash Algorithm 256-bit	2–4
URLLC	Ultra-Reliable Low-Latency Communications	1

1 Introduction

Digital Twins (DTs) have become a standard component in modern industrial automation for connecting physical assets with their virtual counterparts. By maintaining accurate virtual copies of systems like industrial robots, DTs enable real-time monitoring, predictive maintenance, and remote control. In time-critical Cyber-Physical Systems (CPS) applications, the utility of a DT is strictly defined by the synchronization accuracy between the physical robot and the digital model [7]. Consequently, ensuring state consistency typically demands Ultra-Reliable Low-Latency Communications (URLLC) capabilities to support high-frequency updates.

However, meeting these requirements via standard periodic transmission is resource-intensive. Since the DT often possesses a precise estimate of the robot’s state, continuously transmitting raw telemetry data creates significant redundancy. This inefficiency consumes scarce bandwidth without providing new information gain.

To mitigate this redundancy, recent research proposes *identification-based Identification (ID)* synchronization [4]. Unlike classical Shannon-type communication, which aims to reconstruct the full message, ID schemes focus on a verification task: confirming whether the remote state matches a local hypothesis. Building on this concept, prior works [5] demonstrate that transmitting a compressed “verifier” (e.g., a short hash or randomized code) instead of the full state significantly reduces communication traffic. This approach effectively supports goal-oriented communication [3] in low-latency scenarios.

However, implementing ID synchronization in real-time control loops reveals two critical engineering trade-offs. First, regarding latency, the computational overhead of encoding on edge devices may exceed the time saved by reducing transmission traffic. This shift from a communication-bound to a computation-bound bottleneck necessitates a rigorous end-to-end feasibility analysis. Second, reliability becomes a primary concern when moving from exact to semantic verification. In practical control, enforcing bitwise synchronization (standard ID) is often overly conservative, triggering unnecessary updates for benign deviations such as sensor noise. To address this, we extend identification to semantic “safe ranges”, where the system verifies set membership rather than a unique value. This relaxation, however, introduces collision risks, specifically Type II (miss) errors, where dangerous desynchronizations are incorrectly classified as safe. Therefore, quantifying the relationship between the semantic acceptance range and safety risks is essential.

To address these limitations, this thesis presents a complementary study to existing ID frameworks, extending the analysis from pure traffic optimization to an integrated view of latency and reliability in K-Identification (K-ID) systems. The main contributions are:

First, we develop an end-to-end latency model combining computational and transmission delays. We compare cryptographic ([Secure Hash Algorithm 256-bit \(SHA-256\)](#)) and algebraic ([Reed-Solomon Identification \(RS-ID\)](#)) encoding schemes to determine the quantitative thresholds (break-even points) where [ID](#) yields a net latency benefit as a function of bandwidth and verifier length.

Second, we formalize the [K-ID](#) problem for robotic monitoring, distinguishing *safe* from *unsafe* desynchronization via semantic safety sets. We derive quantitative bounds on miss error probabilities induced by [K-ID](#) mechanisms.

Third, we conduct a systematic evaluation using a reproducible software stack. Through Monte Carlo experiments on Gaussian robot trajectories, we expose a non-monotonic dependence between the semantic acceptance range (Set B) and error rates. These results provide design guidelines for balancing semantic tolerance with safety detection in edge-based [DTs](#) networks.

Relation to prior work. While [ID](#) rests on a solid information-theoretic foundation, engineering studies have primarily focused on traffic suppression. We extend this line of work by treating latency as a primary metric alongside traffic and by generalizing from single identification to [K-ID](#), which aligns better with real-world [DT](#) workflows. Our results aim to inform design choices regarding *when to use ID/K-ID* and *with which verifier lengths* under realistic bandwidth conditions.

Organization. Chapter [2](#) introduces the background and related work; Chapter [3](#) details the system model and measurement protocol; Chapter [4](#) describes the software stack and reproducibility; Chapter [5](#) presents the experimental results, including latency–traffic curves parameterized by bandwidth and load; Chapter [6](#) concludes and outlines future engineering directions for [K-ID](#).

2 Background and Related Work

This chapter reviews the theoretical transition from classical data transmission to identification-based synchronization and analyzes the encoding mechanisms required for its implementation. We focus specifically on the mathematical necessity of randomization to overcome structural blindness in finite-field compression. Beyond exact verification, we further examine the generalization of this paradigm to [K-ID](#), illustrating how relaxing strict equality constraints to semantic safety sets allows the synchronization protocol to accommodate the natural variability inherent in physical control loops.

2.1 From Data Reconstruction to Identification

In time-critical control loops, the utility of a [DT](#) depends strictly on its synchronization with the physical asset. Conventional protocols treat this as a data reconstruction problem, where the robot periodically transmits its full state vector S_t (e.g., joint angles and velocities) to the [DT](#). While robust, this approach ignores a critical redundancy: the [DT](#), utilizing its internal kinematic model, already possesses a high-fidelity prediction $\hat{S}_t$ of the robot's state. The communication task, therefore, is not to inform the [DT](#) "where the robot is," but rather to verify "whether the [DT](#)'s prediction is correct."

This shift in objective allows for a fundamental change in coding strategy, moving from Shannon's transmission capacity to the Identification ([ID](#)) capacity defined by Ahlswede and Dueck [1]. In transmission, every possible state must be mapped to a disjoint decoding set, limiting the number of distinguishable states to an exponential growth 2^{nR} relative to the blocklength n . In contrast, identification allows decoding sets to overlap, provided the intersection between any two sets is statistically negligible. This relaxation enables the number of identifiable states to grow doubly exponentially ($2^{2^{nR}}$), implying that a verifier of extremely short length can distinguish a practically infinite number of robot states, provided the system tolerates a vanishingly small error probability.

2.2 Encoding Mechanisms and Structural Bias

To realize this theoretical capacity in practical edge computing, the encoding process must map a high-dimensional state space $\mathcal{S}$ (e.g., 64-byte float vectors) into a low-dimensional verifier space $\mathcal{V}$ (e.g., 8-bit integers). We examine two primary implementations: algebraic encoding ([RS-ID](#)) and cryptographic hashing ([SHA-256](#)), and analyze their inherent risks of systematic bias.

Algebraic and Cryptographic Implementations. The [RS-ID](#) scheme treats the robot's state vector S as coefficients of a polynomial $P_S(x)$ over a finite field $\mathbb{F}_q$. The encoding process evaluates this polynomial at a specific point r , yielding the verifier $v = P_S(r) \pmod{q}$. Geometrically, this represents the state as a curve, and the verifier as a single sample point on that curve. Alternatively, the [SHA-256](#) approach approximates this behavior using cryptographic diffusion. The state is serialized and processed through the [SHA-256](#) algorithm, with the output truncated to the target bit-length n_{ver} . While [RS-ID](#) offers rigorous algebraic bounds on collision probabilities, truncated [SHA-256](#) benefits from hardware acceleration on modern processors, making it computationally favorable for high-frequency control loops.

The Problem of Structural Blindness. A critical limitation of deterministic compression—whether via polynomial evaluation or hashing—is the existence of "structural blindness" or systematic bias. Consider a simplified case where the encoder uses a modulo operation (e.g., $v = S \pmod{M}$). If the deviation between the physical robot and the [DT](#), denoted as error $e = S - \hat{S}$, happens to be a multiple of the modulus M , the generated verifiers will be identical ($S \equiv \hat{S} \pmod{M}$).

Even if the robot's state S changes continuously over time, this specific error pattern e might persist (e.g., a constant sensor offset or a parallel drift). In a deterministic system, the verifiers would consistently match despite the desynchronization, leading to a persistent Type II (Miss) error. The system becomes "blind" to this specific class of errors, rendering the safety check ineffective regardless of the robot's movement.

Randomization as a Countermeasure. To eliminate structural blindness, identification theory mandates the use of *Common Randomness*—a shared seed r (or salt) that changes dynamically with each transmission. In [RS-ID](#), changing r corresponds to sampling the polynomial $P_S(x)$ at a different X-coordinate. In [SHA-256](#), appending a time-variant salt ($S||r$) alters the avalanche dynamics of the hash function. By varying r , the system fundamentally changes the mapping logic at every time step. Although a collision may occur at a specific seed r_1 for a specific error e , the probability that the same error e causes a collision at a new seed r_2 is independent and uniformly bounded. This converts potential "permanent system failures" into "transient probabilistic events," which can be rapidly detected in subsequent time slots.

2.3 Theoretical Limits of Compression

While the introduction of randomization allows for drastic compression, the output length cannot be reduced arbitrarily. The fundamental lower bound for the verifier length L is governed by the algebraic properties of the state representation.

In the polynomial model of [RS-ID](#), a state vector of length k (interpreted as a polynomial of degree $k - 1$) can intersect with another state's polynomial at most at $k - 1$ points. To ensure reliable identification, the size of the evaluation field $Q = 2^L$ must be significantly larger than the number of potential intersection points. If $Q < k$, the "collision points" could saturate the field, making it impossible to find a seed r that distinguishes the two states.

Therefore, a necessary condition for reliability is $2^L > k$, or logarithmically:

$$L > \log_2 k \tag{2.1}$$

This inequality represents the physical manifestation of the double exponential limit. It dictates that the verifier length L does not need to scale linearly with the data size k (as in

transmission), but must only scale logarithmically. As long as the verifier space is large enough to accommodate the "roots" of the difference polynomial, the randomized ID scheme maintains a bounded error probability, enabling secure and efficient synchronization for DTs.

2.4 Semantic Verification: Adapting to Control Flexibility

While standard identification provides a rigorous method for exact state verification, its binary nature—accepting only a single, precise value—often contradicts the operational flexibility required in modern robotic control. In many practical scenarios, the DT does not need to enforce a strict equality constraint ($x = \hat{x}$) but rather needs to guarantee that the robot operates within a permissible semantic range. This nuance necessitates the adoption of K-ID, which generalizes the verification target from a unique point to a semantic safety set.

Consider a practical example of a mobile robot navigating a warehouse. The control policy might specify a safe velocity corridor, such as "maintain speed between 40 cm/s and 60 cm/s." If the robot moves at 45 cm/s while the DT predicted 50 cm/s, a standard ID scheme would flag this as a collision (mismatch) because the hashed values of "45" and "50" are distinct. This triggers an unnecessary data transmission to correct a behavior that is, semantically speaking, perfectly valid. K-ID resolves this inefficiency by defining the acceptance region $\mathcal{B}$ as the interval $[40, 60]$. As long as the robot's state falls within this set, the K-ID verifier confirms validity without consuming bandwidth for minor deviations.

This logic extends beyond simple velocity constraints to more complex spatial and interactive scenarios. For instance, in Human-Robot Collaboration (HRC), the exact coordinates of a robotic arm are less critical than ensuring the end-effector remains outside a defined "human safety bubble." Similarly, for high-precision assembly tasks involving compliant mechanisms, the robot may exhibit minor high-frequency vibrations (jitter) that are mechanically acceptable. Standard ID would treat these micro-vibrations as errors, whereas K-ID can be tuned to ignore high-frequency noise while still detecting significant trajectory deviations.

By shifting the verification paradigm from data exactness to semantic compliance, K-ID aligns the synchronization protocol with the actual control objectives. It allows the system to filter out benign variability—whether it be speed fluctuations, sensor noise, or compliant jitter—and focus limited communication resources on detecting genuine safety violations. This thesis specifically investigates the implementation of such semantic verification using randomized encoding on edge constraints.

3 Design Methodology

This chapter formulates the system model for identification-based synchronization and derives the theoretical bounds for its reliability and latency. We contrast the data flow of classical transmission with the proposed identification scheme, explicitly breaking down the operational stages from state generation to synchronization. We then quantify the collision probabilities and end-to-end latency based on these distinct workflows.

3.1 System Model and Data Flow

We consider a networked control system where a Robot (Sender) synchronizes its state S_t with a Digital Twin (Receiver) maintaining a prediction $\hat{S}_t$. The fundamental difference between the two approaches lies in their data processing pipelines.

3.1.1 Baseline: Traditional Transmission Flow

The standard synchronization protocol follows a linear, “always-on” approach. As illustrated in Figure 3.1, the data flow consists of three sequential stages:

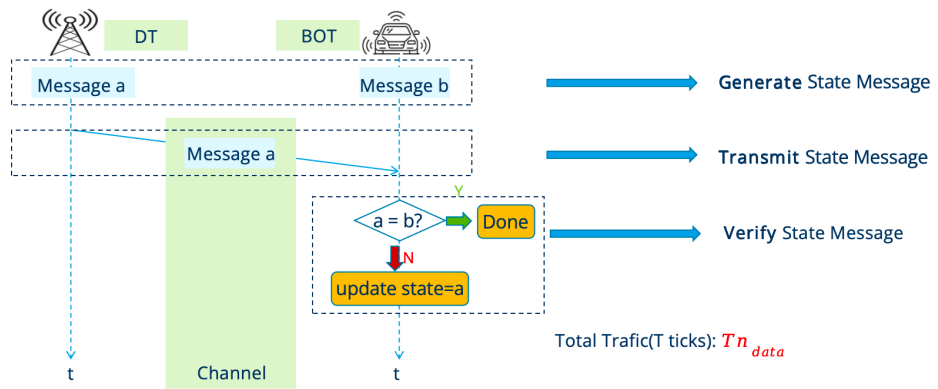


Figure 3.1: Data flow of Traditional Synchronization: The full state is transmitted regardless of the Digital Twin’s knowledge.

1. **Generate:** The robot samples its sensors to generate the full state message S_t of size n_{data} .

2. **Transmit:** The entire payload S_t is transmitted over the channel. This step consumes bandwidth proportional to the state dimension, regardless of whether S_t has changed significantly or matches the DT's prediction.
3. **Verify/Update:** The DT receives the message, overwrites its local prediction $\hat{S}_t$ with the received S_t , and confirms the update.

This process is robust but inefficient, as the **Transmit** stage imposes a heavy, constant latency penalty ($T_{trans} = n_{data}/B$).

3.1.2 Proposed: Identification (ID) Data Flow

The identification-based protocol introduces computational intelligence to reduce communication load. As shown in Figure 3.2, the pipeline is expanded to five stages, introducing a “Verify-then-Correct” logic:

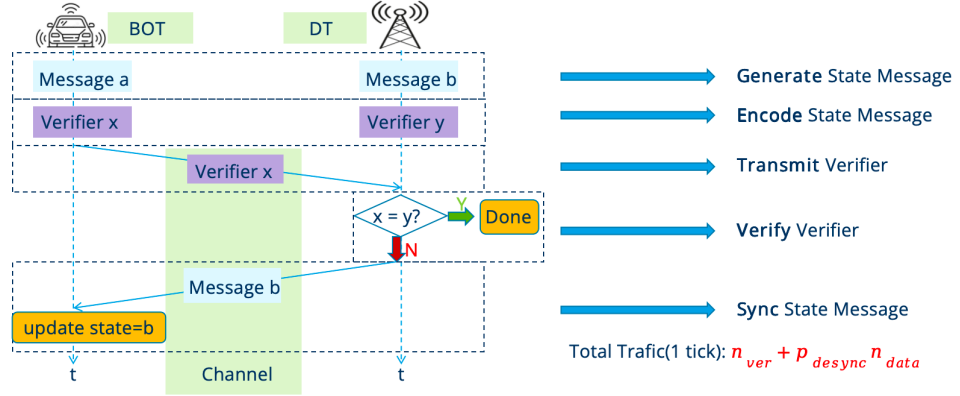


Figure 3.2: Data flow of ID Synchronization: High-bandwidth transmission is replaced by local encoding and conditional fallback.

1. **Generate:** Both the Robot and the DT independently generate their respective state views: the actual state S_t and the predicted state $\hat{S}_t$.
2. **Encode:** Instead of preparing the full message for transmission, the Robot computes a compact signature (verifier) $v = E(S_t, r)$ using a shared seed r . Simultaneously, the DT computes its local reference $\hat{v} = E(\hat{S}_t, r)$. This step introduces a computational latency (t_{ver}) but drastically compresses the data.
3. **Transmit:** Only the short verifier v (size $n_{ver} \ll n_{data}$) is sent over the channel.
4. **Verify:** The DT compares the received v with its local $\hat{v}$.
5. **Sync (Conditional):**
 - If $v = \hat{v}$, the process terminates (Done).
 - If $v \neq \hat{v}$, a mismatch is detected. The system enters a fallback mode, requesting the full transmission of S_t to resynchronize.

3.2 Encoding Mechanisms and Reliability Analysis

The safety of the ID flow depends on the reliability of the **Verify** stage. A critical failure occurs if the states differ ($S_t \neq \hat{S}_t$) but their verifiers collide ($v = \hat{v}$), defined as a Type II Error (p_{miss}). We investigate two distinct encoding mechanisms, detailing their construction and theoretical error bounds.

3.2.1 Algebraic Construction and Error Analysis (RS-ID)

The core challenge of the RS-ID scheme is to map a high-dimensional state vector S of length n_{data} into a single element within a finite field $\mathbb{F}_q$, where the field size $q = 2^{n_{ver}}$ is determined by the output bandwidth constraint.

Polynomial Construction (The “Chunking” Process) Since the operations are defined over $\mathbb{F}_q$, a state vector S typically exceeds the capacity of a single field element (i.e., $n_{data} \gg n_{ver}$). To resolve this dimensionality mismatch, we decompose the bitstream S into k segments, where each segment represents a coefficient in the field:

$$k = \left\lceil \frac{n_{data}}{n_{ver}} \right\rceil \quad (3.1)$$

Let S be represented as a vector of coefficients $\{s_0, s_1, \dots, s_{k-1}\}$, where each $s_i \in \mathbb{F}_q$. We then construct a polynomial $P_S(x)$ of degree at most $k - 1$:

$$P_S(x) = \sum_{i=0}^{k-1} s_i x^i \quad (3.2)$$

The verifier v is obtained by evaluating this polynomial at a shared random seed r : $v = P_S(r)$.

Proof of Collision Probability A Type II error (Miss) occurs when the Digital Twin predicts an incorrect state $\hat{S} \neq S$, yet the generated verifiers are identical ($P_{\hat{S}}(r) = P_S(r)$). We define a *difference polynomial* $\Delta(x)$:

$$\Delta(x) = P_S(x) - P_{\hat{S}}(x) \quad (3.3)$$

Since $S \neq \hat{S}$, at least one coefficient differs, making $\Delta(x)$ a non-zero polynomial with a degree $\deg(\Delta) \leq k - 1$. A collision event is mathematically equivalent to finding a root of this difference polynomial: $\Delta(r) = 0$.

We invoke the **Schwartz-Zippel Lemma** (Lemma 1 in [6]), which states that a non-zero polynomial of degree d over a field $\mathbb{F}_q$ has at most d roots. Mapping this theorem to our system parameters:

- The polynomial degree d corresponds to the chunk count $k \approx n_{data}/n_{ver}$.
- The sample space size corresponds to the field size $q = 2^{n_{ver}}$.

Consequently, for a uniformly distributed random seed r , the probability of collision is bounded by:

$$P_{miss}^{RS} = \Pr[\Delta(r) = 0] \leq \frac{\deg(\Delta)}{q} < \frac{k}{2^{n_{ver}}} = \frac{\lceil n_{data}/n_{ver} \rceil}{2^{n_{ver}}} \quad (3.4)$$

Theoretical Implication This derivation highlights the structural trade-off in algebraic identification. The term k in the numerator represents the complexity of the data: as the

data size n_{data} grows, the polynomial becomes more complex (higher degree), increasing the number of potential roots (collisions). Therefore, to maintain a constant safety margin P_{miss} , the verifier length n_{ver} must scale logarithmically with the data size.

3.2.2 Cryptographic Construction and Error Analysis (SHA-256)

Unlike the algebraic structure of RS-ID, the security of hash-based identification relies on computational hardness assumptions. We verify the state using a truncated cryptographic hash function (SHA-256):

$$v = \text{Truncate}(\text{SHA-256}(S||r), n_{ver}) \quad (3.5)$$

where $||$ denotes concatenation with the random seed r .

Random Oracle Assumption To analyze the collision probability theoretically, we adopt the **Random Oracle Model (ROM)** paradigm introduced by Bellare and Rogaway [2]. In this model, the hash function $H(\cdot)$ is idealized as a mathematical oracle that maps every unique input query to a response selected **uniformly and independently** from the output domain $\{0, 1\}^{n_{ver}}$.

Proof of Collision Probability (Derivation from Uniformity) Under the ROM, for any robot state S , the hash output $H(S||r)$ is a random variable following a uniform distribution over the verifier space of size $N = 2^{n_{ver}}$. Given a pre-calculated verifier $\hat{v}$ from the Digital Twin, the probability of a collision (Type II error) is equivalent to the probability that a uniformly selected random value hits a specific target in the sample space.

$$P_{miss}^{SHA} = \Pr[H(S||r) = \hat{v}] = \frac{1}{|\{0, 1\}^{n_{ver}}|} = \frac{1}{2^{n_{ver}}} \quad (3.6)$$

This derivation relies on the property that in a uniform distribution, the probability of any single outcome is the reciprocal of the sample space size.

Comparative Analysis: RS-ID vs. SHA-256 A critical distinction arises when comparing this bound with Equation (3.4) (the RS-ID bound).

- **RS-ID Dependency:** The error probability in RS-ID scales linearly with the data size ($k \approx n_{data}/n_{ver}$), reflecting the geometric constraint that higher-degree polynomials have more roots.
- **SHA-256 Independence:** In contrast, under the Random Oracle assumption, the error probability for SHA-256 is **invariant** to the input data size n_{data} .

This property makes the cryptographic approach structurally more robust for large-scale state vectors (e.g., LiDAR point clouds), as the “collision risk” does not accumulate with data length. However, this comes at the cost of losing the unconditional algebraic guarantee provided by the Schwartz-Zippel Lemma, replacing it with a computational assumption.

3.2.3 Tolerance-Based Verification (K-ID)

The ID data flow described in Section 3.1 assumes a strict equality check between a single predicted state $\hat{S}_t$ and the true state S_t . In practice, however, many cyber-physical systems can tolerate bounded deviations without requiring a full resynchronization. We therefore extend the verification stage to a *tolerance-based* rule, referred to as **K-ID**.

Acceptable Set: Given the DT prediction $\hat{S}_t$ and a tolerance radius K , the DT defines an acceptable set

$$\mathcal{A}_K(\hat{S}_t) = \{s \mid \|s - \hat{S}_t\| \leq K\}. \quad (3.7)$$

In the experiments in this thesis, we use a scalar state for clarity (so $\|\cdot\|$ is absolute value), but the K-ID logic generalizes to vector-valued states with an appropriate norm.

K-ID Decision Rule: The Robot sends a truncated hash verifier

$$v = \text{Truncate}(\text{SHA256}(S_t \| r), n_{\text{ver}}). \quad (3.8)$$

The DT pre-computes the set of valid verifiers for the acceptable states,

$$\mathcal{V}_K = \{\text{Truncate}(\text{SHA256}(s \| r), n_{\text{ver}}) \mid s \in \mathcal{A}_K(\hat{S}_t)\}, \quad (3.9)$$

and *accepts* if $v \in \mathcal{V}_K$. Otherwise, it triggers fallback and requests the full payload.

Safety Miss Rate: Define an *unsafe* event as the true state lying outside the tolerance region, i.e., $S_t \notin \mathcal{A}_K(\hat{S}_t)$. A miss occurs when the DT accepts despite being unsafe:

$$p_{\text{miss},K} = \Pr[v \in \mathcal{V}_K \mid S_t \notin \mathcal{A}_K(\hat{S}_t)]. \quad (3.10)$$

Under the Random Oracle assumption used in Section 3.2.2, the verifier v for an unsafe state is uniform over $\{0, 1\}^{n_{\text{ver}}}$. Hence, the conditional miss probability is bounded by the fraction of verifier space covered by acceptable hashes:

$$p_{\text{miss},K}^{\text{SHA}} \approx \frac{|\mathcal{V}_K|}{2^{n_{\text{ver}}}} \leq \frac{|\mathcal{A}_K(\hat{S}_t)|}{2^{n_{\text{ver}}}}. \quad (3.11)$$

This highlights a key trade-off: increasing K enlarges the acceptable set (reducing fallback frequency), but also increases the miss probability conditional on being unsafe.

Unsafe Rate and Latency: Let $p_{\text{unsafe}}(K) = \Pr[S_t \notin \mathcal{A}_K(\hat{S}_t)]$. If the prediction error $e = S_t - \hat{S}_t$ is modeled as $\mathcal{N}(0, \sigma^2)$, then

$$p_{\text{unsafe}}(K) = \Pr[|e| > K] = 2(1 - \Phi(K/\sigma)). \quad (3.12)$$

Since fallback is triggered when the DT is unsafe and *does not* miss, the expected latency follows the same structure as Section 3.3 but with the tolerance-aware terms:

$$T_{KID} = t_{\text{ver}} + \frac{n_{\text{ver}}}{B} + p_{\text{unsafe}}(K)(1 - p_{\text{miss},K})\frac{n_{\text{data}}}{B}. \quad (3.13)$$

The empirical sweep in Chapter 5 instantiates this model and visualizes the safety-latency trade-off across K and n_{ver} .

3.3 End-to-End Latency Model

Based on the data flows defined in Section 3.1, we mathematically formulate the expected latency.

Traditional Latency (T_{trad}): Since the traditional flow always executes the full transmission, its latency is deterministic:

$$T_{trad} = \frac{n_{data}}{B} \quad (3.14)$$

where B is the channel bandwidth.

Identification Latency (T_{ID}): The ID flow is probabilistic. The total time includes the mandatory encoding and verifier transmission, plus the conditional penalty for fallback. Matching the stages in Figure 3.2:

$$T_{ID} = \underbrace{t_{ver}}_{\text{Encode}} + \underbrace{\frac{n_{ver}}{B}}_{\text{Transmit Verifier}} + \underbrace{p_{desync}(1 - p_{miss})\frac{n_{data}}{B}}_{\text{Sync (Fallback Penalty)}} \quad (3.15)$$

- t_{ver} : The computational time for the **Encode** stage.
- p_{desync} : The probability that the DT's prediction fails ($S_t \neq \hat{S}_t$).
- $(1 - p_{miss})\frac{n_{data}}{B}$: The penalty is incurred only when a desynchronization exists AND is successfully detected by the verifier check.

This model highlights the fundamental trade-off: ID reduces the transmission term from $\frac{n_{data}}{B}$ to $\frac{n_{ver}}{B}$ but adds a computational cost t_{ver} . Therefore, ID is advantageous only when the bandwidth B is limited (making transmission expensive) or when the prediction accuracy is high (keeping p_{desync} small).

4 Implementation

4.1 Experimental Parameters

This chapter details the practical setup used to validate the theoretical models from Chapter 3. For the bandwidth and p_{desync} sensitivity plots in Chapter 5, all experiments are configured with a desynchronization probability of $p_{desync} = 0.1$. We evaluate two data payloads: a small 96-bit message and a larger 4001-bit message. For each payload, we test verifier lengths of $n_{ver} \in \{4, 16\}$ bits, representing a small and a large verifier size.

4.2 Theoretical Collision Probabilities

From the formulas derived in Section 3.2, we can pre-calculate the expected theoretical collision probabilities (p_{miss}) for our chosen parameters.

4.3 Latency Plot Generation

To visualize the implications of the hybrid latency model, we generate latency curves parameterized by bandwidth using the Python scripts in the `experiments/` folder.

4.3.1 Parameters

All latency plots in Chapter 5 use the experimental parameter grid $n_{data} \in \{96, 4001\}$ bits and $n_{ver} \in \{4, 16\}$ bits with $p_{desync} = 0.1$. The compute overheads t_{ver} are taken from empirical measurements (Table 5.1), and the collision probabilities p_{miss} are computed from the theoretical bounds (Table 5.2).

4.3.2 Logic

For each configuration (n_{data}, n_{ver}) and each system (SHA-256 vs. RS-ID), the script evaluates the expected latency

$$L_{ID}(B) = t_{ver} + \frac{n_{ver}}{B} + p_{desync} (1 - p_{miss}) \frac{n_{data}}{B} \quad (4.1)$$

over a log-spaced bandwidth grid $B \in [B_{min}, B_{max}]$. The traditional baseline is computed as $L_{trad}(B) = n_{data}/B$. The resulting curves are plotted on log-log axes to reveal the break-even regime where ID-based synchronization becomes beneficial.

The generated figures are shown in Chapter 5 for $n_{data} = 96$ bits (Figure 5.1) and for $n_{data} = 4001$ bits (Figure 5.2).

In addition, we visualize the sensitivity to desynchronization by sweeping $p_{desync} \in [0, 1]$ at a fixed bandwidth and plotting L_{ID} against p_{desync} for the same parameter grid (Figures 5.3 and 5.4).

4.3.3 SHA-256

The collision probability for a truncated hash function under the Random Oracle Model is independent of the data size and is given by Equation 3.6:

$$P_{miss}^{SHA} = 2^{-n_{ver}} \quad (4.2)$$

The corresponding numerical values for our selected verifier lengths are reported in Table 5.2 in Chapter 5.

4.3.4 RS-ID

The collision probability for the algebraic construction is bounded by the Schwartz-Zippel Lemma, as shown in Equation 3.4. This bound depends on both data size (n_{data}) and verifier length (n_{ver}).

$$P_{miss}^{RS} \leq \frac{\lceil n_{data,bits}/n_{ver} \rceil}{2^{n_{ver}}} \quad (4.3)$$

The computed values for $n_{data} \in \{96, 4001\}$ bits and $n_{ver} \in \{4, 16\}$ bits (with the bound capped at 1 when applicable) are reported in Table 5.2 in Chapter 5. These values highlight how the theoretical safety of RS-ID degrades with larger data sizes for a fixed verifier length, a key distinction from the cryptographic approach.

4.4 K-ID Parameter Sweep Setup

In addition to the bit-based latency evaluation, we implement the tolerance-based K-ID model introduced in Section 3.2.3 and perform a parameter sweep over (K, n_{ver}) to quantify the safety-latency trade-off of the K-ID decision rule.

4.4.1 Model and Metrics

The K-ID sweep uses a truncated SHA-256 verifier, and reports the following metrics:

- **Miss rate (conditional):** $\Pr[\text{accept} \mid \text{unsafe}]$ as defined in Equation (3.11).
- **Fallback rate:** $\Pr[\text{fallback}]$ over all ticks.
- **Expected latency:** the hybrid model in Equation (3.13).
- **Speedup:** ratio of traditional transmission time (n_{data}/B) to the expected K-ID latency.

4.4.2 State Space and Quantization

To avoid an artificially small discrete domain (which can force the empirical miss rate to become exactly zero for moderate n_{ver}), the robot state is modeled as a continuous random variable and then quantized:

1. Sample $S_t \sim \mathcal{N}(\mu, \sigma^2)$, clip to a physical range $[S_{min}, S_{max}]$.
2. Quantize to resolution Δ (default $\Delta = 0.1$), and hash the resulting integer index.

This yields a discrete domain size of approximately $(S_{max} - S_{min})/\Delta + 1$.

4.4.3 Parameter Grid

Unless otherwise stated, the sweep uses:

- Payload size: $n_{data} = 96$ bits.
- Bandwidth: $B = 1$ Mbps.
- Compute overhead: $t_{ver} = 2.13 \mu s$ (SHA-256, 96 bits; Table 5.1).
- State distribution: $\mu = 50$, $\sigma = 12$, clipped to $[0, 100]$.
- Quantization: $\Delta = 0.1$.
- Sweep: $n_{ver} \in \{4, 8, 12, 16\}$ bits and $K \in \{2, 5, 10, 20\}$.
- Monte Carlo ticks: 200,000 per configuration, seed = 42.

4.4.4 Reproducibility

The sweep and plot generation are implemented in the `experiments/` folder:

- `experiments/kid_parameter_sweep.py`: runs the simulation and optionally writes a CSV.
- `experiments/plot_kid_parameter_sweep.py`: reads the CSV and produces the figures used in Chapter 5.

The following commands reproduce the CSV and plots (paths are relative to the repository root):

```
python3 experiments/kid_parameter_sweep.py \
  --n-data-bits 96 --bandwidth 1e6 --ticks 200000 --seed 42 \
  --robot-mean 50 --robot-std 12 --physical-min 0 --physical-max 100 \
  --value-resolution 0.1 --t-compute-us 2.13 \
  --nver 4 8 12 16 --k 2 5 10 20 \
  --out-csv experiments/results/kid_sweep.csv

python3 experiments/plot_kid_parameter_sweep.py \
  --csv experiments/results/kid_sweep.csv --out-dir experiments/results
```

5 Results and Evaluation

5.1 Empirical Parameter Verification

To instantiate the theoretical models derived in Chapter 3, we first characterize the system parameters experimentally. This includes measuring the computational overhead (t_{ver}) and calculating the theoretical collision probabilities (p_{miss}).

5.1.1 Encoding Latency (t_{ver})

The computational overhead of the encoding step, t_{ver} , was measured empirically by running 100,000 `send` operations for each system configuration. In addition to the sample mean, we estimate a **95% confidence interval (CI)** for the mean using a batch timing method (100 timing batches, each covering an equal share of operations).

Interpretation of Table 5.1: The table reports the *sample mean* $\bar{t}_{ver}$ in microseconds. The CI quantifies measurement uncertainty due to runtime noise (OS scheduling, cache effects, Python overhead). In our measurements, the 95% CI half-widths are below $1\ \mu s$ for all configurations. Therefore, the reported means are sufficiently stable for the latency-model comparisons in this chapter.

Note that the **mean** values can shift slightly across runs and machines (Python runtime, CPU frequency scaling, background load). For reproducibility and consistency with the figures in this chapter, we keep the table mean values aligned with the parameter set used for the plotted latency evaluation, and we report representative 95% CI half-widths to quantify the measurement uncertainty.

Analysis of t_{ver} Results: The measurement results in Table 5.1 reveal distinct performance characteristics between the algebraic (RS-ID) and cryptographic (SHA-256) approaches:

1. **Stability of SHA-256:** Consistent with the theoretical discussion in Section 3.2, the computation time of SHA-256 is largely insensitive to the output verifier length. For the short payload ($n_{data} = 96$ bits), the latency remains nearly constant ($\sim 2.2\ \mu s$) when truncating to either $n_{ver} = 4$ or $n_{ver} = 16$ bits. When increasing the payload to $n_{data} = 4001$ bits, the latency rises to $\sim 15\ \mu s$, which is expected due to the block-based processing of SHA.

Table 5.1: Measured Verification Time t_{ver} (mean $\pm$ 95% CI half-width, with relative half-width)

System (n_{data})	n_{ver} (bits)	
	4	16
SHA-256		
96 bits	2.13 ± 0.04 (1.9%)	2.24 ± 0.01 (0.4%)
4001 bits	14.65 ± 0.31 (2.1%)	15.41 ± 0.09 (0.6%)
RS-ID		
96 bits	14.55 ± 0.39 (2.7%)	3.79 ± 0.02 (0.5%)
4001 bits	51.78 ± 0.94 (1.8%)	38.57 ± 0.36 (0.9%)

2. **The Chunking Effect in RS-ID:** A counter-intuitive phenomenon is observed for RS-ID: for the small payload ($n_{data} = 96$ bits), increasing the verifier length from $n_{ver} = 4$ to $n_{ver} = 16$ bits *reduces* the measured latency substantially (from $14.55 \mu s$ to $3.79 \mu s$). This can be explained by the polynomial construction mechanism. RS-ID represents the input as k coefficients, where $k = \lceil n_{data}/n_{ver} \rceil$. With $n_{ver} = 4$, the data is sliced into 24 chunks, requiring more iterations of field operations. With $n_{ver} = 16$, the chunk count drops to 6. For small datasets, the reduction in loop iterations outweighs the increased per-operation cost of arithmetic in a larger field.

5.1.2 Theoretical Collision Probability (p_{miss})

Based on the bounds derived in Chapter 3 (Equations (3.6) and (3.4)), we compute the theoretical collision probability p_{miss} for the parameter grid used in the latency model. Note that for RS-ID, if the polynomial degree exceeds the field size, the bound is capped at 1.0.

Table 5.2: Theoretical Collision Probability (p_{miss}) for Bit-Based Parameters

System (n_{data})	n_{ver} (bits)	
	4	16
SHA-256 (any n_{data})	6.25×10^{-2}	1.53×10^{-5}
RS-ID		
96 bits	1.00	9.16×10^{-5}
4001 bits	1.00	3.83×10^{-3}

5.2 End-to-End Latency Evaluation

Using the hybrid latency model and the calibrated parameters from Section 5.1, we analyze the system performance across varying network conditions and prediction qualities.

5.2.1 Sensitivity to Channel Bandwidth

We first evaluate the expected latency as a function of channel bandwidth (B). Figures 5.1 and 5.2 compare the Traditional approach against the ID-based schemes.

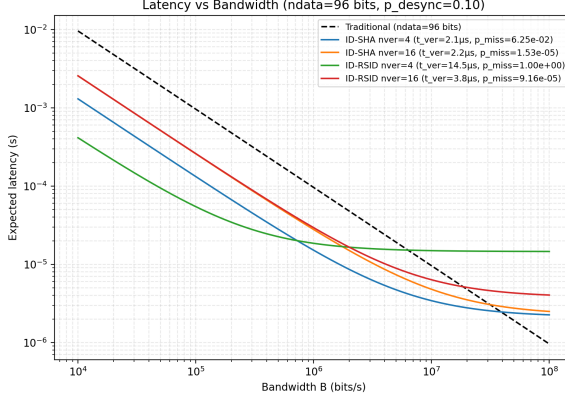


Figure 5.1: Expected latency vs. bandwidth for $n_{data} = 96$ bits ($p_{desync} = 0.1$).

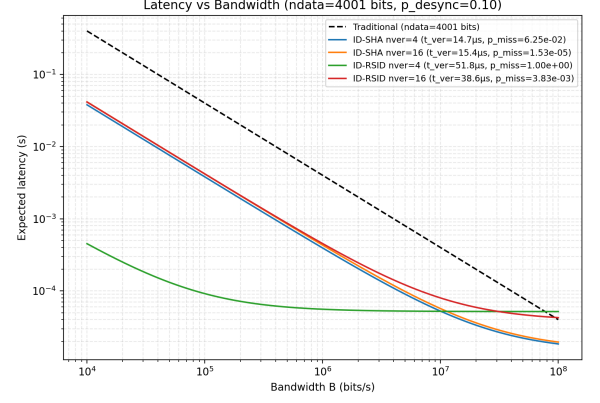


Figure 5.2: Expected latency vs. bandwidth for $n_{data} = 4001$ bits ($p_{desync} = 0.1$).

Discussion (Bandwidth Trade-off): The results illustrate the dominance of the Identification scheme in bandwidth-constrained environments.

- **Low Bandwidth Regime ($B < 1$ Mbps):** The transmission time of raw data dominates the Traditional latency. The ID scheme offers an order-of-magnitude reduction by transmitting only the compact verifier.
- **High Bandwidth Regime ($B > 10$ Mbps):** As transmission becomes instantaneous, the latency floor is determined by the computational overhead t_{ver} . Consequently, the Traditional scheme (which has zero encoding overhead) eventually outperforms the ID schemes, creating a clear "Cross-over Point."

5.2.2 Sensitivity to Prediction Quality

Next, we examine the impact of the Digital Twin's prediction accuracy (p_{desync}) at a fixed bandwidth ($B = 5$ Mbps).

Discussion (Prediction Quality):

- **Break-even Analysis:** For the short payload (Fig. 5.3), the ID-SHA curves intersect the Traditional line at approximately $p_{desync} \approx 0.8$. This implies that ID is beneficial as long as the Digital Twin's prediction accuracy exceeds 20%. For the large payload (Fig. 5.4), the transmission penalty is so high that ID-SHA remains superior across nearly the entire range of p_{desync} .
- **Anomaly Analysis (RS-ID $n_{ver} = 4$):** A notable anomaly appears in the RS-ID results for $n_{ver} = 4$ bits (orange line in Fig. 5.3), which appears flat and low. This is a **theoretical artifact** caused by the saturation of collision probability ($p_{miss} = 1.0$, see Table 5.2). Mathematically, this sets the fallback penalty term to zero. In reality, this

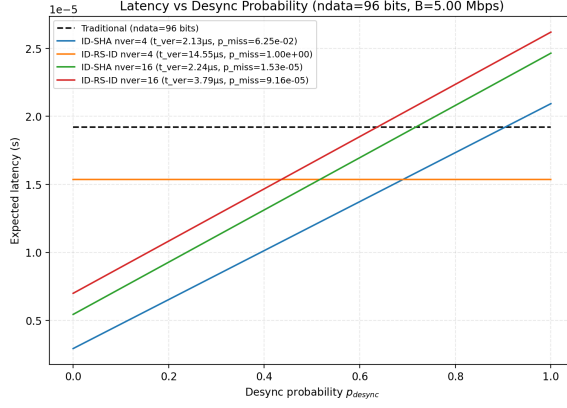


Figure 5.3: Expected latency vs. p_{desync} for $n_{data} = 96$ bits ($B = 5$ Mbps).

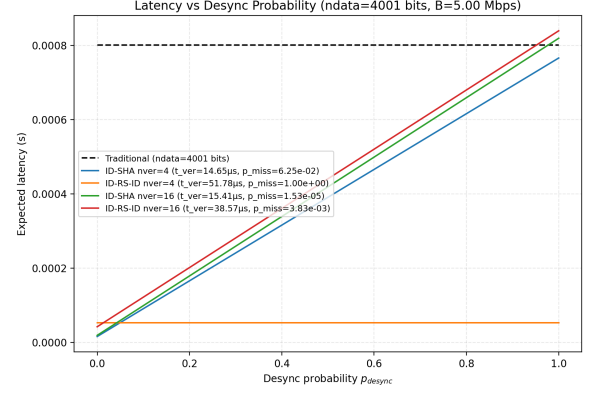


Figure 5.4: Expected latency vs. p_{desync} for $n_{data} = 4001$ bits ($B = 5$ Mbps).

represents a catastrophic failure where desynchronization is never detected. Therefore, this configuration is invalid for safety-critical systems despite its apparent low latency.

5.3 K-ID Safety-Latency Trade-off

This section evaluates the tolerance-based K-ID model introduced in Section 3.2.3 and implemented as described in Section 4.4. The experiment sweeps $n_{ver} \in \{4, 8, 12, 16\}$ and $K \in \{2, 5, 10, 20\}$ for a 96-bit payload at $B = 1$ Mbps.

5.3.1 Sweep Results

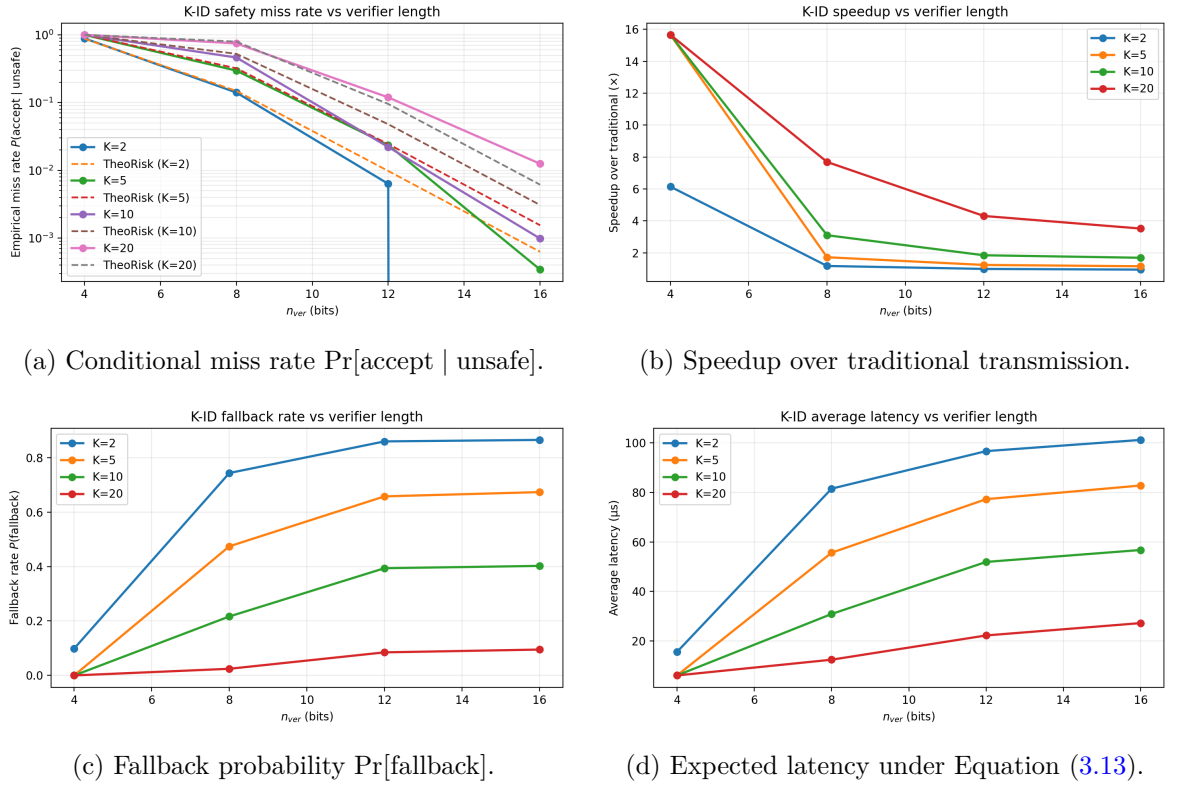
Figure 5.5 summarizes the main metrics. The miss rate is plotted as the conditional safety risk $\Pr[\text{accept} \mid \text{unsafe}]$, i.e., the probability that an unsafe state is incorrectly accepted due to a verifier collision with the acceptable set.

Analysis (Effect of n_{ver}): Increasing n_{ver} reduces the conditional miss rate approximately exponentially, consistent with the Random Oracle scaling in Equation (3.11). However, a larger n_{ver} also increases the mandatory verifier transmission term n_{ver}/B and, more importantly, reduces the chance of “accidental acceptance” when unsafe, which increases detection and therefore increases fallback frequency. As a result, higher n_{ver} improves safety but can reduce speedup (and increase latency) depending on how often the system enters unsafe states.

Analysis (Effect of K): Increasing K enlarges the acceptable set. This has two opposing effects that are visible in Figure 5.5:

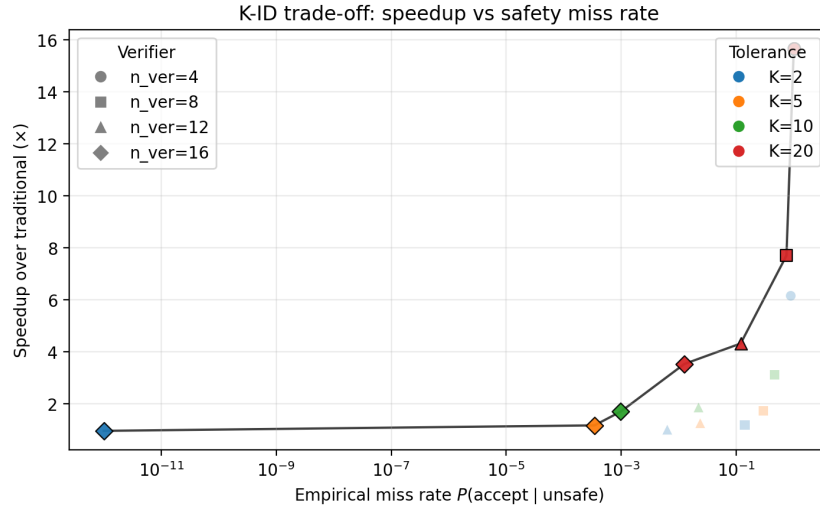
- The conditional miss rate $\Pr[\text{accept} \mid \text{unsafe}]$ increases with K , because more acceptable states mean more opportunities for a hash collision with an unsafe verifier.
- The fallback rate and latency decrease with K because the system is unsafe less often (fewer ticks lie outside the tolerance interval), so the expensive full-payload fallback is triggered less frequently.

This demonstrates the fundamental K-ID trade-off: a larger tolerance improves performance and reduces fallback, but raises the conditional risk of accepting an unsafe state.


 Figure 5.5: K-ID parameter sweep results for $n_{data} = 96$ bits at $B = 1$ Mbps.

5.3.2 Pareto Frontier Interpretation

To make the trade-off explicit, Figure 5.6 plots speedup against the *conditional* miss rate $P(\text{accept} | \text{unsafe})$ and highlights the non-dominated (Pareto-optimal) configurations.


 Figure 5.6: Pareto view of the K-ID trade-off: speedup vs. conditional miss rate $P(\text{accept} | \text{unsafe})$.

Overall Risk vs. Conditional Miss Rate: The x-axis in Figure 5.6 is a *conditional* safety metric: it describes how likely an unsafe state is accepted *given that* the system is already unsafe. To assess end-to-end operational risk, it is often more informative to consider the *overall* probability of an unsafe acceptance per tick,

$$P(\text{unsafe} \cap \text{accept}) = P(\text{unsafe}) \cdot P(\text{accept} \mid \text{unsafe}). \quad (5.1)$$

This overall risk accounts for how frequently unsafe states occur (which depends strongly on K and the prediction error distribution). Figure 5.7 therefore replots the trade-off using $P(\text{unsafe} \cap \text{accept})$ on the x-axis.

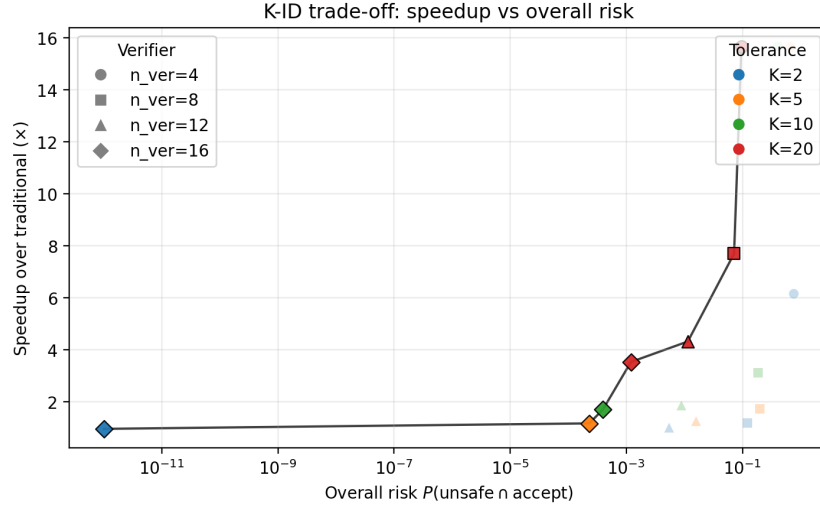


Figure 5.7: Pareto view using overall risk $P(\text{unsafe} \cap \text{accept})$: speedup vs. total unsafe-accept probability per tick.

Discussion: The solid curve in Figure 5.6 connects the Pareto-optimal configurations (the *frontier*). These points represent operating choices where improving safety (moving left, lower conditional miss rate) necessarily requires sacrificing speedup (moving down), and vice versa. Points in the lower-right region are *dominated* (simultaneously slower and less safe) and can be discarded.

In deployment, the choice of (K, n_{ver}) should be driven by an application-level safety budget on $p_{\text{miss}, K}$ (or an overall risk measure that also weights the unsafe frequency) together with a latency target. The sweep provides a practical menu of operating points to select from based on these constraints.

5.3.3 RS-ID Comparison and Uniform Error Model

The baseline K-ID sweep above uses a truncated SHA-256 verifier and a Gaussian prediction error. To compare against the algebraic verifier family, we rerun the sweep with an **RS-ID** verifier (single-tag, tag position 2) and replace the error model with a **uniform** distribution centered at the mean. The uniform half-width is chosen to match the variance of the Gaussian case ($a = \sqrt{3} \sigma$).

Figure 5.8 shows the resulting metrics side-by-side for `sha256_trunc` and `rsid`. The overall trends remain consistent with the K-ID model: increasing n_{ver} reduces collision-driven acceptance, while increasing K reduces fallback frequency (by making unsafe states rarer), at the cost of enlarging the acceptable set.

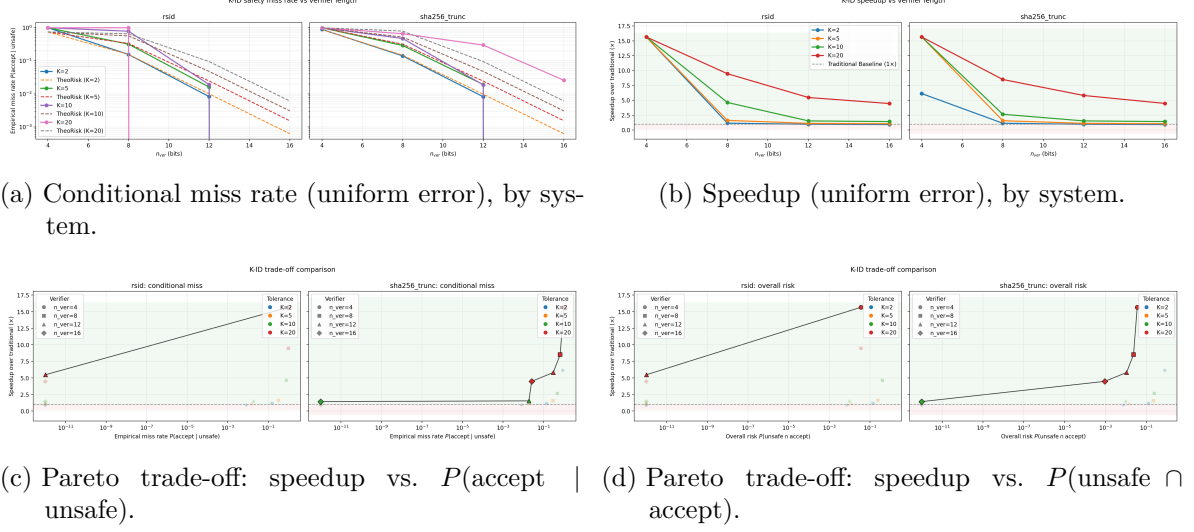


Figure 5.8: K-ID sweep comparison of truncated SHA-256 vs. RS-ID under a uniform prediction error model ($n_{data} = 96$ bits, $B = 1$ Mbps).

6 Conclusion and Future Work

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A Appendix

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